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Subject: IS547- Foundations of Data Curation

Final Project Report

1. Scenario, Use Case & Curation Goals

a. Project Scenario

For this project, I worked with a Dryad dataset associated with the research study “**A Study of Impulsivity and Adverse Childhood Experiences in a Population Health Setting.**” The dataset contains information related to adverse childhood experiences (ACEs), impulsivity measures, and participant-related variables used in the published study.

In this project, I took the role of a research data curator working within a scholarly data-sharing environment. Instead of performing new statistical analysis, the main focus of the project was to evaluate how well the dataset has been prepared for sharing, interpretation, transparency, and future reuse. Since the dataset is publicly available through the Dryad repository, the project focused on understanding whether another researcher could realistically interpret and reuse the data without needing direct clarification from the original authors.

The project followed a CURATE(D)-based approach discussed throughout the course. As part of this workflow, I reviewed the repository record, metadata, documentation, dataset structure, associated article, and supporting materials to evaluate how effectively the dataset supports discoverability, provenance, reproducibility, and long-term reuse.

A major part of the project involved examining the relationship between the deposited dataset and the associated research article. During the review process, it became clear that the article plays an important role in helping users interpret the dataset, understand the study context, and identify how certain variables and measures were used within the research workflow.

This project also connects to broader data curation concepts discussed in class, including metadata quality, documentation practices, provenance tracking, ethical data sharing, transparency, reproducibility, and preservation-oriented repository support.

b. Use Case

The primary use case for this project was to evaluate whether the dataset is sufficiently understandable and reusable for secondary researchers working outside the original study team.

To assess this, I reviewed:

- the Dryad repository record,
- the dataset files,
- metadata and documentation,
- and the associated research article.

The project focused on identifying whether enough contextual and methodological information exists for another researcher to:

- understand the variables,
- interpret the dataset correctly,
- connect the data to the published findings,
- and potentially reproduce parts of the study workflow or analysis process.

Another important part of the project was evaluating how effectively the repository supports transparency, discoverability, citation, and long-term access through persistent identifiers and structured repository metadata.

c. Curation Goals

The main goals of this project were:

1. To evaluate whether the dataset is understandable and reusable for independent researchers.
2. To assess the quality and completeness of the dataset's metadata and supporting documentation.
3. To identify areas where interpretation of the dataset depends heavily on the associated research article.
4. To examine whether provenance and methodological details are sufficiently documented for transparency and reproducibility.
5. To evaluate how effectively the Dryad repository supports discoverability, citation, preservation, and long-term access.
6. To identify documentation or metadata gaps that may affect reuse or interpretation.
7. To review ethical and responsible data-sharing considerations related to sensitive population health data.
8. To develop curator-focused recommendations that could improve future reuse, transparency, and reproducibility of the dataset.

These goals align closely with the broader data curation concepts discussed throughout the course, particularly metadata quality, provenance, transparency, reproducibility, FAIR-oriented practices, and responsible research data management.

2. Dataset & Artifact Characterization

The dataset selected for this project is **“Data from: A Study of Impulsivity and Adverse Childhood Experiences in a Population Health Setting,”** deposited in the **Dryad** repository with the DOI **10.5061/dryad.66t1g1k8x**. The dataset supports a published population health study examining the relationship between **adverse childhood experiences (ACEs), impulsivity, and health-related outcomes**.

The dataset is organized as participant-level tabular data, where each row represents an individual participant record and each column represents a study variable or measurement. The associated study included approximately **17,482 participants**. The variables relate to areas such as ACE exposure, impulsivity measures, demographic information, and health-related indicators used in the published research.

a. Dataset Context and Research Area

This dataset belongs to the field of **population health and behavioral research**. Because it deals with adverse childhood experiences and impulsivity, the data has a sensitive human-subject context. This makes curation especially important because secondary researchers need clear documentation to interpret variables responsibly and avoid misusing or misunderstanding the data.

In this type of research, simply sharing a dataset is not enough. Users also need enough context to understand how the data was collected, what the variables mean, how scores were created, and how the dataset connects to the published study. For this reason, the associated article was treated as a key artifact in the curation process.

b. Dataset Structure and Repository Contents

The Dryad deposit includes the dataset file, repository-level metadata, citation information, DOI, and a link to the associated article. The dataset is structured in rows and columns, which makes it suitable for review as a research data file and for long-term repository access.

During my review, I examined the repository record, dataset structure, metadata fields, article connection, and available documentation. The repository provides useful discovery and citation information, but the dataset's full interpretation appears to depend partly on the associated article, especially for understanding ACE-related variables, impulsivity measures, and methodological context.

c. Curation Artifacts and Supporting Materials

For the final submission, I created the main CURATE(D) artifacts expected for this project: a final curation report, curation log, metadata improvement notes, and draft author communication/recommendations. I also included supporting review notes from the progress stage, such as the repository review, metadata assessment, and article-dataset comparison. Together, these artifacts show how the dataset was evaluated and how the final recommendations were developed.

3. Ethical, Legal & Policy Compliance

a. Human Subjects and Privacy

This dataset is based on population health research involving adverse childhood experiences (ACEs), impulsivity, and behavioral health-related information. Because ACE-related data can reflect sensitive personal experiences, participant privacy is one of the most important ethical concerns in this project.

As part of the curation review, I assessed whether the dataset appeared to be shared in a de-identified form and whether the repository materials avoid exposing direct personal identifiers. Even when data is publicly available in Dryad, sensitive health-related data still requires careful interpretation and responsible reuse. For this reason, the curation assessment considered whether secondary users would have enough context to understand the data without misusing or over-interpreting it.

b. Repository and Policy Compliance

The dataset is deposited in Dryad, which supports scholarly data sharing through persistent DOI assignment, repository metadata, citation information, and long-term access. These features help the dataset meet common expectations for research data sharing, discoverability, and preservation.

From a curation perspective, the Dryad deposit supports basic compliance because the dataset is publicly accessible, citable, and connected to an associated research article. However, repository access alone does not guarantee full reuse. For responsible reuse, users also need clear documentation, variable descriptions, and methodological context.

c. Responsible Reuse and Interpretation

A key ethical issue in this project is not only whether the data is accessible, but whether it can be interpreted responsibly by someone outside the original research team. Because the dataset involves ACEs and impulsivity measures, unclear documentation could lead to misunderstanding of sensitive behavioral or health-related variables.

During the review, I found that the associated article is important for understanding the research context and interpreting the dataset correctly. This means that ethical reuse depends on reading the dataset and article together. In my recommendations, I treat stronger documentation and clearer variable-level explanation as ethical improvements, not only technical improvements, because they help future users interpret the data more carefully and responsibly.

4. Curation Workflow

a. Week 1 – Dataset and Article Review

The first stage of the workflow involved carefully reviewing the Dryad dataset and the associated research article. During this phase, I examined the repository structure, DOI information, metadata fields, and overall dataset organization.

I identified major variables related to:

- adverse childhood experiences (ACEs),
- impulsivity measures,
- demographic information,
- and population health indicators.

At this stage, I also began maintaining a curation log to document observations, decisions, and issues identified during the review process.

b. Week 2 – Documentation and Metadata Assessment

The second stage focused on assessing documentation quality and metadata completeness. I reviewed:

- descriptive metadata,
- dataset naming conventions,
- repository metadata,
- variable descriptions,
- and available supporting documentation.

I also compared the dataset structure with the methodology described in the article to evaluate whether the repository materials provided enough information for independent understanding and reuse.

One important observation during this stage was that interpretation of several variables still depends heavily on the associated publication.

Workflow Decisions:

- Treated the associated article as a central curation artifact
- Prioritized metadata review and documentation clarity
- Recorded documentation gaps and reuse concerns in the curation notes

c. Week 3 – Reproducibility and Transparency Review

The third stage involved evaluating reproducibility and transparency by reviewing whether the publicly shared dataset appeared sufficient to support the analyses and conclusions discussed in the article.

This phase included identifying:

- missing contextual explanations,
- undocumented transformations,
- dependencies on the publication for interpretation,
- and areas where reproducibility may be limited.

The workflow during this stage focused more on evaluation and assessment rather than modifying the original dataset.

Workflow Decisions:

- Compared dataset variables with published study methods
- Evaluated whether an independent researcher could reasonably interpret the dataset
- Documented reproducibility limitations and transparency concerns

d. Week 4 – Ethical, Privacy, and FAIR Assessment

The fourth stage focused on ethical, privacy, and FAIR-related review. Since the dataset contains sensitive ACE-related population health information, I assessed the repository materials from a responsible reuse perspective.

This stage included:

- reviewing participant privacy considerations,
- assessing metadata completeness,
- evaluating repository transparency,
- and examining FAIR alignment related to findability, accessibility, and reusability.

Workflow Decisions:

- Considered ethical interpretation as part of curation quality
- Evaluated whether documentation supports responsible secondary reuse
- Included ethical observations in the final recommendations

e. Week 5 – Final Reporting and Artifact Preparation

The final stage involved consolidating all findings, observations, and recommendations into the final curation report and supporting artifacts. During this phase, I finalized the curation log, refined recommendations, and organized project materials for submission and GitHub documentation.

The final workflow artifacts include:

- metadata assessment notes,
- repository review findings,
- article-dataset comparison notes,
- reproducibility observations,
- and final curation recommendations.

Overall, the workflow followed a structured CURATE(D)-style evaluation process focused on understanding how effectively the dataset supports transparency, documentation quality, reproducibility, discoverability, and responsible reuse.

5. Metadata & Data Documentation

a. Repository Metadata

The dataset repository in Dryad includes core descriptive metadata such as the dataset title, authors, publication year, DOI, abstract, funding information, keywords, and citation details. The dataset is linked directly to the associated research article, which improves discoverability and allows users to connect the deposited files with the published study. The DOI assigned through Dryad supports persistent access and long-term referencing of the dataset.

The repository also includes file-level information and download access for the deposited CSV files. From a discovery perspective, the metadata is sufficient for locating and citing the dataset. However, additional contextual metadata would improve independent understanding and secondary reuse by researchers unfamiliar with the original study domain.

b. Dataset Documentation

The dataset contains structured CSV files with observational and measurement data related to adverse childhood experiences, impulsivity measures, and demographic indicators used in the associated research study. The files are organized in tabular format and can be opened using common analytical tools such as Excel, Python, R, or statistical software packages.

During the curation process, I reviewed the dataset structure, column organization, naming conventions, and file relationships. I identified that several variables could be understood only by referring back to the journal article, which creates a dependency on external documentation. While the repository includes basic documentation, some variables and scoring approaches lack detailed explanations needed for independent reuse.

To improve interpretability, I documented important metadata elements including:

- dataset title and DOI,
- file names and formats,
- number of rows and columns,
- key variables,
- variable groupings,
- relationships between files,
- and the connection between the dataset and the published article.

This documentation supports better understanding of the dataset structure and helps future researchers navigate the files more efficiently.

c. Metadata Quality Assessment

As part of the curation workflow, I evaluated the metadata quality using principles related to discoverability, interpretability, and reuse. The repository performs well in terms of accessibility because the files are openly available through Dryad and supported by a persistent identifier. The dataset can also be cited properly through the provided citation information.

However, the assessment identified several metadata limitations affecting reuse:

- some variable names are abbreviated or unclear,
- scoring calculations are not fully documented,
- preprocessing or transformation steps are not explicitly described,
- and relationships between analytical methods and deposited files require interpretation from the journal article.

These gaps reduce the ability of an independent researcher to fully understand the dataset without consulting external materials. From a data curation perspective, this impacts transparency and reproducibility because important contextual knowledge remains implicit rather than documented directly within the repository.

d. Data Dictionary and Supporting Documentation

To address documentation gaps identified during the assessment, I created supplementary curation artifacts including:

- a metadata assessment document,
- dataset structure documentation,
- repository review notes,
- article-to-dataset comparison records,
- and a curation log documenting observations and decisions made throughout the project.

These artifacts function as supporting documentation intended to improve dataset interpretation and reuse. The documentation also records where metadata was incomplete, where clarification was required from the article, and where additional contextual explanation would benefit future users.

The curation log additionally provides transparency into the review process by documenting decisions, observations, and identified limitations throughout the evaluation workflow.

e. Recommended Metadata Improvements

Based on the curation assessment, I identified several metadata and documentation improvements that would strengthen long-term reuse and reproducibility of the dataset:

- Add a more detailed variable-level data dictionary explaining abbreviations, scoring methods, and derived variables.
- Include clearer descriptions for demographic and impulsivity-related variables to reduce ambiguity for secondary users.
- Provide documentation connecting statistical methods described in the article with the deposited dataset files.
- Include workflow notes describing preprocessing, cleaning, or transformation activities performed before publication.
- Expand README documentation with examples explaining relationships between dataset files and variables.
- Add more contextual metadata directly within the repository so researchers can understand the dataset without depending entirely on the associated publication.

These recommendations support independent understanding, transparency, discoverability, and responsible long-term reuse of the dataset, which are central goals within the data curation process.

6. Provenance & Data Lineage

a. Dataset Origin and Source Lineage

The dataset curated in this project originated from a publicly available Dryad repository associated with a peer-reviewed research article examining adverse childhood experiences, impulsivity, and related behavioral indicators. The repository includes deposited dataset files, citation information, DOI assignment, and direct links to the associated publication describing the study methodology and findings.

As part of the provenance assessment, I traced the relationship between the journal article and the deposited dataset files to understand how the shared data supports the published research outcomes. This included reviewing dataset structure, variable organization, file relationships, and methodological explanations provided within the article.

The dataset represents a post-publication stage in the research data lifecycle, meaning that data collection and analysis had already been completed before repository deposition. Therefore, this project focused primarily on evaluating the transparency, traceability, and documentation of the shared research outputs.

b. Provenance Documentation and Curation Tracking

Throughout the project, I maintained documentation of curation activities and observations through multiple supporting artifacts. These included:

- repository review documentation,
- metadata assessment notes,
- article and dataset comparison records,
- documentation gap analysis,
- and a chronological curation log.

The curation log was used to record workflow steps, decisions, identified limitations, interpretation challenges, and recommendations generated during the evaluation process. Maintaining these records improved transparency by documenting how conclusions were reached throughout the project.

This documentation also supports traceability because future reviewers can understand which parts of the dataset were assessed, what issues were identified, and how the curation workflow was conducted.

c. Assessment of Transformations and Analytical Lineage

One important part of provenance evaluation involved examining whether the analytical processes described in the article could be connected clearly to the deposited dataset files. During this review, I compared variables discussed in the publication with the available dataset columns, file structures, and documented metadata.

The assessment indicated that several analytical transformations and preparation steps likely occurred before publication, but detailed processing documentation was not fully included within the repository. In some cases, interpretation of variables depended heavily on explanations found in the associated article rather than metadata available directly in the deposited files.

I also observed that some variables appeared to represent processed or derived measures rather than raw observations. However, the repository did not provide detailed workflow documentation explaining how these derived variables were generated. This creates partial lineage gaps because future users may not fully understand how the dataset moved from original collection to final analytical form.

d. Actions and Transformations Reviewed During Curation

As part of the provenance assessment, several curation and evaluation actions were performed to better understand the dataset lineage and workflow transparency.

These actions included:

- reviewing relationships between deposited files,
- identifying connections between dataset variables and article findings,
- examining variable naming conventions and coded values,
- comparing statistical measures described in the publication with the deposited data,
- evaluating whether scoring methods and analytical variables were sufficiently documented,
- identifying potential preprocessing or derived-variable creation steps,
- and documenting metadata gaps or unclear transformations within the curation log.

Through this process, I determined that the dataset supports partial analytical transparency but lacks complete transformation lineage documentation. Additional workflow records describing preprocessing, statistical preparation, and derived-variable generation would significantly improve provenance traceability and independent reuse.

e. Transparency and Lineage Traceability

To improve provenance transparency during the curation process, I documented:

- the source repository and DOI,
- dataset acquisition details,
- relationships between deposited files,
- metadata observations,
- documentation limitations,
- and connections between the article methodology and the deposited dataset.

These records establish a clearer lineage trail for the dataset and support future reuse by making the curation process more explicit and reviewable.

The inclusion of curated documentation and workflow tracking also supports accountability within the research data lifecycle by ensuring that assessment decisions, identified limitations, and curation recommendations remain transparent for future researchers and reviewers.

7. Reproducibility & Transparency

I conducted a reproducibility-oriented curation assessment. The goal was to evaluate whether the deposited dataset, repository metadata, supporting documentation, and associated article provide enough information for an independent researcher to understand and evaluate the study without contacting the original authors.

a. Reproducibility-Oriented Assessment

I reviewed the Dryad dataset files, repository record, metadata, and associated article together to see whether the research workflow could be understood from the shared materials. The dataset supports partial reproducibility because it is publicly accessible, has a DOI, is linked to the article,

and contains participant-level variables related to ACEs, impulsivity, demographics, and health-related indicators.

However, full reproducibility is limited because some details are not fully documented in the repository itself. These include variable-level explanations, scoring procedures, preprocessing steps, derived variables, and how certain measures connect to the reported findings in the article. Because of this, the article is necessary for understanding parts of the dataset.

b. Transparency of Curation Process

To make my own curation process transparent, I documented the steps and decisions made throughout the project. I created and maintained artifacts such as the curation log, repository review, metadata assessment, article-dataset comparison, documentation gap notes, and final recommendations.

These materials show what I reviewed, what issues I identified, and how my conclusions were developed. This allows the instructor or reviewers to inspect the intermediate artifacts and understand the reasoning behind my assessment.

c. Key Limitations Identified

The main limitations affecting reproducibility and transparency were:

- limited variable-level documentation,
- unclear scoring or derived-variable explanations,
- lack of detailed preprocessing or transformation notes,
- dependence on the associated article for interpretation,
- and limited standalone documentation in the repository.

These issues do not make the dataset unusable, but they reduce how easily another researcher could independently understand, evaluate, or reuse the dataset.

d. Recommendations

To improve reproducibility and transparency, I recommend adding a clearer data dictionary, expanded README documentation, explicit scoring explanations, and notes describing preprocessing or transformation steps. These improvements would help future users inspect the dataset, understand key variables, and reuse the data more responsibly without relying too heavily on the article alone.

8. Archiving & Preservation

a. Repository Selection and Long-Term Access

The dataset used in this project was deposited in the Dryad repository, which is a widely recognized research data repository designed to support long-term preservation and public access to scholarly datasets. Dryad provides persistent identifiers through DOI assignment, stable hosting,

citation support, and open access distribution, all of which strengthen long-term discoverability and reuse.

The use of Dryad is appropriate for this dataset because it supports preservation of research outputs associated with published journal articles and allows future researchers to access the files without institutional restrictions.

b. File Formats and Preservation Suitability

The deposited dataset files are stored in CSV format, which is a non-proprietary and preservation-friendly file format commonly recommended for long-term research data access. CSV files are widely supported across analytical software platforms and reduce the risk of future accessibility issues caused by software obsolescence.

The repository also includes descriptive metadata and citation information that support future interpretation and retrieval of the dataset.

However, while the file formats themselves are suitable for preservation, long-term usability also depends on documentation quality. In this project, I identified that additional contextual documentation would improve preservation of interpretability alongside preservation of the files themselves.

c. Preservation Challenges Identified

During the curation assessment, I observed that some important contextual information exists primarily within the associated article rather than directly inside the repository documentation. This creates a preservation risk because future users may struggle to fully interpret variables, scoring methods, or analytical context if external references become unavailable or difficult to access.

The main preservation-related limitations identified were:

- limited variable-level explanations,
- incomplete transformation documentation,
- dependence on the article for interpretation,
- and minimal standalone repository documentation.

These issues affect preservation of meaning and usability even though the files themselves are technically preserved.

d. Preservation Recommendations

To strengthen long-term preservation and reuse, I recommend:

- expanding the README documentation,
- adding a detailed data dictionary,
- documenting preprocessing and scoring procedures,
- and including more contextual metadata directly within the repository.

These additions would improve not only file preservation, but also preservation of understanding, transparency, and future interpretability of the dataset.

e. Preservation of Curation Artifacts

As part of this project, the curation artifacts created during the assessment process were organized and preserved within the GitHub repository. These artifacts include the curation log, metadata assessment, repository review, article-dataset comparison, preliminary findings, and final evaluation documents.

Maintaining these artifacts supports transparency and provides additional preservation evidence by documenting the assessment workflow, identified limitations, and recommendations developed throughout the project.

9. Findings & Recommendations

a. Major Findings

The curation assessment showed that the dataset is valuable, well organized, and publicly accessible through a trusted repository. The Dryad deposit includes persistent identification through a DOI, open access availability, and a direct connection to the associated journal article, which supports discoverability and citation.

The dataset structure was generally consistent and interpretable, and the deposited files could be connected to the study objectives discussed in the article. The repository also used preservation-friendly file formats such as CSV, which support long-term accessibility and reuse.

However, the assessment also identified several documentation and transparency limitations that affect independent understanding and reproducibility. Important contextual information, including variable explanations, scoring methods, preprocessing details, and transformation steps, was not always fully documented within the repository itself. In many cases, understanding the dataset required referring back to the associated article.

The project therefore found that the dataset supports partial independent reuse, but additional documentation would significantly improve transparency, interpretability, and long-term usability.

b. Metadata and Documentation Findings

The metadata provided by Dryad supports dataset discovery and citation, but variable-level documentation remains limited. Some variable names and coded values were unclear without article context, and the repository lacked a detailed standalone data dictionary explaining all analytical variables and scoring approaches.

The review also identified limited explanation of preprocessing or derived-variable creation, making it difficult to fully trace analytical lineage from data collection to published findings.

c. Reproducibility Findings

From a reproducibility perspective, the repository provides enough information to understand the general research purpose and dataset structure. However, full reproducibility is limited because important workflow details and transformation steps are not completely documented.

This demonstrates that public availability alone does not guarantee reproducibility. Effective reuse also depends on clear metadata, transparent documentation, and sufficient contextual explanation for independent researchers.

d. Recommendations

Based on the curation assessment, the following recommendations were identified:

- expand the README documentation with clearer dataset explanations,
- create a detailed variable-level data dictionary,
- document preprocessing, scoring, and transformation steps more explicitly,
- improve standalone repository documentation so interpretation does not depend heavily on the journal article,
- and strengthen metadata descriptions for better independent understanding and reuse.

Implementing these improvements would support stronger transparency, reproducibility, long-term preservation, and responsible secondary use of the dataset.

10. Conclusion

This project evaluated a Dryad dataset related to adverse childhood experiences, impulsivity, and population health from a CURATE(D)-based data curation perspective. The focus was not on new analysis, but on assessing whether the dataset is understandable, transparent, reproducible, and reusable for future researchers.

Overall, the dataset is valuable, publicly accessible, and supported by Dryad's DOI-based preservation and citation system. However, the review also showed that public access alone is not enough for strong reuse. Some variable explanations, scoring details, preprocessing steps, and transformation information need clearer documentation.

A major finding was that the associated article is essential for interpreting the dataset, which means the repository documentation could be strengthened to support more independent reuse. Through metadata review, provenance assessment, reproducibility evaluation, and curation logging, this project showed how data curation improves transparency, trust, and long-term usability of shared research data.

GitHub Link: https://github.com/Bhagvesha-P/Foundations_of_Data_Curation

11. References

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GenAI Declaration

I used generative AI tools for brainstorming, grammar refinement, formatting support, and improving the organization of the report. All curation decisions, interpretations, findings, and evaluations presented in this project were reviewed, modified, and finalized by me based on my own understanding of the dataset and course requirements.